

Revision Summary — Trading-Aware Agents in Sugarscape

Response to *Computational Economics* Reviewer #1 (Major Revision)

This document summarizes every change made to the manuscript in response to the reviewer's eight points, plus the additional improvements made during revision. Section references are to the revised manuscript (Manuscript_v2.tex).

Part 1 — Point-by-point response to Reviewer #1

Point 1: "The framing suggests 'learning trading strategies' and 'jointly optimizing movement and exchange,' but the algorithmic design is movement-only learning + exogenous trading."

The reviewer was correct. The manuscript claimed joint optimization of movement and trade, while the implementation learns movement only, with trade executed by the classical deterministic Rule T. This was the root cause of Points 1, 2, and 4 together.

What changed. The entire contribution was reframed around the accurate concept of **trade-aware movement**: a learned movement policy conditioned on neighbors' marginal rates of substitution (MRS), with bilateral exchange unchanged. Every passage claiming joint optimization was rewritten:

- Abstract: now states explicitly that "the policy learns *where to be*, not *how to bargain*," and that Rule T executes exchange.
- Introduction: "unifies movement and trade" replaced by "couples movement decisions to local market information"; the deliberate narrowing of scope is stated and justified.
- Sections 2.1–2.4: all "learns trading strategies" / "implicit trading decisions" language corrected.
- Conclusion: rewritten to match.
- A full-document sweep confirmed zero surviving instances of "jointly optimize," "joint optimization," "unifies movement and trade," or "learn trading strategies."

New text added. A "**Scope of Learning**" paragraph now opens Section 3.2, stating in plain terms: the DRL policy governs movement only; Rule T (price = geometric mean of the two parties' MRS) executes all trades; the agent has no policy over what, how much, at what price, or whether to trade; the term "trade-aware" is retained because the observation includes market state and the learned behavior responds to it.

Point 2: "The differentiator is not 'RL in Sugarscape' but 'movement learning conditioned on trading-related state while trade remains rule-based.'"

We adopted the reviewer's own characterization as the paper's stated contribution. The introduction now presents a threefold contribution: (1) a conceptual update to Sugarscape that embeds bounded-rational adaptation *without replacing the classical trading mechanism*; (2) a methodologically controlled comparison between Rule-M and PPO-learned movement under matched conditions; (3) a computational laboratory for policy

experiments. Section 2.2 repositions the novelty against prior RL-in-Sugarscape work (Brearcliffe & Crooks 2020; Jäger 2021) precisely as the reviewer framed it: those works learned movement over resource state only; ours embeds *market information* (neighbors' MRS) into the observation so movement can respond to trading opportunity.

Point 3: "Baseline and DRL regimes differ in multiple structural parameters simultaneously, making causal attribution to RL invalid."

This was a presentation error, not a design error — and the fix makes that unambiguous. The original Table 1 compared "Baseline" (meaning the original Epstein–Axtell 1996 configuration, shown only for reference) against "DRL extension" (meaning the settings of this study). The two-column format made it appear that rule-based and DRL agents were run under different parameters. In fact, **both conditions in every experiment share identical environmental parameters; the only difference is the movement policy.**

What changed.

- The two-column table was **deleted** and replaced by (a) a short reference paragraph describing the canonical 1996 configuration and the seven deliberate departures from it, each with its justification, and (b) a **new single-column Table 1** listing this study's parameters, whose caption states in bold: "All values apply identically to rule-based and DRL conditions; the only difference between conditions is the movement policy."
- An explicit **identity-of-conditions statement** was added in Section 3.1 (bold in the text) listing every held-fixed parameter.
- The statement is **echoed in Section 4.1** immediately before the headline carrying-capacity numbers, so the causal attribution is established at the point of claim.

Point 4: "'Agents learn trading strategies by observing neighbours' MRS' and 'Trading is not an explicit action' cannot both be true."

Resolved by the same reframe as Point 1. The specific sentence the reviewer quoted (Section 2.2) was rewritten to: agents learn *movement* strategies conditioned on neighbors' MRS, positioning themselves to exploit exchanges executed by Rule T. Section 2.3's claim that "the action space includes both directional moves and implicit trading decisions" was corrected to "the action space comprises directional moves only." The Scope of Learning paragraph eliminates the inconsistency permanently: the manuscript now says one thing, everywhere.

Point 5: "The observation design 'ensures the Markov property' is a misconception in multi-agent settings."

The reviewer was correct. Section 3.3.1 was rewritten (now titled "Decision Problem and Decentralized Learning"):

- The joint system is now correctly identified as a **homogeneous decentralized partially observable Markov decision process (Dec-POMDP)** in the sense of Oliehoek & Amato (2016).
- **Non-stationarity is explicitly acknowledged:** because other agents' policies co-evolve during training, the transition kernel experienced by any single agent is non-

stationary (citing Lowe et al. 2017; Foerster et al. 2018). Parameter sharing is presented as a partial mitigation, not a solution.

- The training regime is described honestly as an **approximate solution by independent PPO under parameter sharing**, with empirical support from Yu et al. (2022, MAPPO).
- The overclaim "ensures the Markov property" was replaced by the precise statement that the observation is "*Markovian for the decentralized policy*," with its exact meaning defined.

Point 6: "The observation equation is unclear and potentially dimensionally incorrect."

The notation $(4v+1) \times 4$ was correct but unexplained; it is now fully specified. Section 3.2.1 now states that the agent observes the **four cardinal arms of its von Neumann neighborhood** (v cells in each of N/S/E/W plus its own cell), yielding $n = 4v+1 = 41$ cells at $v = 10$ — a deliberate "cross-shaped" design mirroring the directional action space (each observed cell lies on a traversable path; unreachable diagonals are excluded). The four channels per cell are enumerated, with the welfare-surplus channel given as an explicit formula. The section specifies that the 41×4 matrix is flattened and concatenated with the 6-element internal vector to form a **170-dimensional input**, removing all dimensional ambiguity.

Point 7: "Reward definition is underspecified and may be conceptually inconsistent."

The original text conflated realized and expected welfare; the rewrite commits to one and labels it honestly. Section 3.2.3 now:

- Gives the reward as a numbered equation with clean notation: the welfare difference between the current state and the candidate destination under a **deterministic forward model** of movement and metabolic consumption, minus a death penalty.
- Names this an **expected-utility heuristic** and states its exact scope: the prediction is exact for the deterministic components (movement, metabolism) and may diverge from realized welfare only through stochastic within-step events (regrowth, trade outcomes).
- Relates the scheme to **potential-based reward shaping** (Ng et al. 1999), including the honest acknowledgment that the formal $\gamma\Phi(s') - \Phi(s)$ form would include a discount factor our welfare-difference omits — negligible at $\gamma = 0.99$ but stated for completeness.
- Formalizes the death penalty as a configurable environment hyperparameter (`c_death`, the `death_multiplier` in the implementation), set to 1 in all experiments, with sensitivity flagged as future work.

Point 8: "Price and 'equilibrium' metrics are not rigorously defined."

The metric is now formally defined, and — importantly — the claims were narrowed to what the metric actually measures. Section 3.5 defines the plotted quantity exactly: the **cumulative geometric mean of executed Rule-T trade prices** per replicate run, with the cross-run mean trajectory and cross-run standard deviation $\sigma(t)$ over $R = 50$ replicates. Two

honesty provisions were added: (a) $\sigma(t)$ measures **cross-replicate consistency** of the price trajectory, *not* within-market price dispersion at a moment in time; (b) a cumulative average contracts mechanically as trades accumulate, so comparisons are only made at matched time points across conditions. Section 4.2 was rewritten accordingly: the claim is that trade-aware movement produces markedly more *consistent* (cross-seed-reliable) price convergence, driven by **encounter frequency** — the only channel through which a movement policy can influence prices when Rule T is identical across conditions. The abstract and hypothesis H2 were aligned with this precise formulation. Within-step dispersion analysis and the Foley (1994) benchmark are noted as future work.

Part 2 — Improvements beyond the reviewer's points

These changes were made proactively during revision because the reviewer's concerns exposed related weaknesses, and because filling in real statistics revealed places where the original prose did not match the data.

Full statistical reporting. A statistical-conventions paragraph now opens Section 4. Every numerical comparison reports cross-replicate mean $\pm$ SD for both conditions, the percent difference with a 95% bootstrap confidence interval (10,000 resamples), the paired t-statistic with degrees of freedom, exact p-values, and Cohen's d. Previously the paper gave bare percentages ("approximately 28%") with a generic " $p < 0.001$."

Corrections where the data contradicted the original prose. Two original claims did not survive contact with the actual numbers, and were corrected rather than papered over:

- *Per-capita wealth*: originally claimed to rise "by about 7%"; the data show it is **11.4% lower** under DRL. Section 4.4 was rewritten to explain the true picture: a substantially larger surviving population shares the gains from improved allocation, so per-capita wealth falls while **total societal wealth rises (+13.0%)**. This is arguably a more interesting economic finding than the original claim.
- *Consume-regrow ratio*: originally "6–7% lower"; the data show **3.4–3.7% lower**. Corrected throughout.

Honest reporting of H4 (inequality). The Gini-based prediction is **not supported**: terminal Gini differs by ≈ 0.002 (deterministic) and is marginally *higher* under DRL on stochastic maps. Rather than hiding this, Section 4.5 states it directly and reports H4 as **partially supported**: the summary Gini is unchanged, but the distributional *shape* prediction holds on deterministic maps — a Kolmogorov–Smirnov test rejects equality of the wealth distributions, and upper-tail quantiles (90th/95th/99th percentile) are systematically lower under DRL (99th percentile: 561.7 vs 685.8), quantifying the "thinner upper tail."

Two new analyses deepening the empirical contribution.

- *Kaplan–Meier survival analysis* (Section 4.1): agent lifetimes reconstructed from per-step records; KM median survival is 45% longer under DRL (16 vs 11 steps deterministic; 26 vs 18 stochastic); log-rank tests reject equality of survival distributions overwhelmingly. This shows the DRL advantage holds across the entire mortality trajectory, not only at episode end.

- *Tail-sensitive inequality measures* (Section 4.5): Theil's T index and top-10% wealth share supplement the Gini (which is least sensitive precisely in fat-tailed regimes; Atkinson 1970). These reveal a genuine nuance: on stochastic maps Theil's T is significantly *higher* under DRL, mechanically traced to the larger share of high-metabolism survivors — an efficiency-equality trade-off reported transparently and propagated consistently to the Section 4 summary, the conclusion, and the abstract.

New Limitations subsection (Section 5) stating four limitations explicitly: (1) only movement is learned, bargaining is not; (2) the expected-utility reward has not been compared head-to-head against a strictly realized-welfare reward; (3) no MRS-channel ablation was run, so the claim that trade-awareness (rather than general spatial competence) drives the results is not yet isolated; (4) parameter-sharing PPO constrains behavioral heterogeneity relative to independent learning (IPPO), flagged as future work.

Consistency and technical cleanup. Cobb–Douglas notation harmonized to upright text throughout; all Unicode em/en-dashes converted to LaTeX conventions; two mis-encoded smart quotes and all curly apostrophes removed (the file is now pure ASCII and compiles identically on any engine); figure subcaptions unified from "RL agents" to "DRL agents"; mislabeled subfigure labels in the spatial figure corrected; a grammatically broken citation fixed; an orphaned one-sentence subsection (5.2) folded into 5.1; hypothesis H2, the abstract, and the conclusion cross-checked against the recalibrated results so no section contradicts another.

References. Eight new entries are required in the bibliography: Oliehoek & Amato (2016); Lowe et al. (2017); Foerster et al. (2018); Yu et al. (2022); Kaplan & Meier (1958); Collett (2015); Atkinson (1970); de Witt et al. (2020). One entry (McCarlie & Hunter 2021) is no longer cited and can be removed.

Part 3 — Open items before resubmission

1. **Verify every new statistic against the raw data.** The Kaplan–Meier medians, log-rank χ^2 values, Theil's T values, top-10% shares, and survival fractions were computed in a separate tool session; each must be re-derived from the replication datasets before submission, since the data are public on Zenodo and any discrepancy would be discoverable.
2. **Add the eight new BibTeX entries** (and spot-check page numbers against the original papers).
3. **Optional but recommended: inference-time MRS ablation.** Evaluating the *frozen* trained policy with the MRS observation channel zeroed (no retraining; ~50 evaluation episodes per condition) would directly test whether the policy uses market information, converting Limitation 3 into a positive validation result. This is the single strongest remaining improvement available at low cost.
4. **Supervisor read of the final PDF.** The revision involved extensive AI-assisted editing across many rounds; a human content read is the appropriate final safeguard before resubmission.